

FlagAlgebraToolbox: A Package for Flag Algebra Computations (by Levente Bodnár)

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Outline

- ▶ Flag algebras and SDP
- ▶ `FlagAlgebraToolbox`
- ▶ Sample results and examples of code

Flag Algebras

- ▶ Large unknown graph G
- ▶ Inequalities between subgraph densities
- ▶ Razborov'07: Flag algebras
 - ▶ Asymptotically true inequalities (as $v(G) \rightarrow \infty$)
 - ▶ Formal rules for deriving them
 - ▶ One aspect: semi-definite programming

Turán Problem via Flag Algebras

- ▶ Turán function:

$$\text{ex}(n, K_3) := \max\{e(G) : v(G) = n, G \not\supseteq K_3\}$$

- ▶ Construction: $T_n^2 := K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$
 - ▶ $\text{ex}(n, K_3) \geq e(T_n^2) = \lfloor n^2/4 \rfloor$

Subgraph Densities

- ▶ K_3 -free G of order $n \rightarrow \infty$
- ▶ $\#F := \#(F, G) = \#$ induced copies of F in G
 - ▶ $e(G) = \#K_2$
- ▶ Density of F :

$$p(F) := p(F, G) = \#F / \binom{n}{v(F)} = \mathbf{Prob}\{G[\text{random } v(F)\text{-set}] \cong F\}$$

Obtaining Inequalities

$$\blacktriangleright A = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix} \succeq 0$$

$$\blacktriangleright \mathbf{v}_x = \left(\frac{d(x)}{n-1}, \frac{\bar{d}(x)}{n-1} \right), \quad x \in V(G)$$

$$\blacktriangleright 0 \leq \mathbf{v}_x A \mathbf{v}_x^T$$

$\blacktriangleright$ **Average** over x (and ignore $o(1)$):

$$\begin{aligned} \frac{1}{n} \sum_x \alpha \frac{d^2(x)}{(n-1)^2} &= \frac{\alpha}{n^3} \sum_x \left(\sum_{y \sim x} 1 \right) \left(\sum_{z \sim x} 1 \right) \\ &= \frac{\alpha}{6 \binom{n}{3}} \times 2 \#P_3 = \frac{\alpha}{3} p(P_3) \end{aligned}$$

$$\blacktriangleright 0 \leq \left(\frac{2\beta}{3} + \frac{\alpha}{3} \right) p(P_3) + \left(\frac{2\beta}{3} + \frac{\gamma}{3} \right) p(\bar{P}_3) + \gamma p(\bar{K}_3)$$

Bounding Edge Density from Above

$$\blacktriangleright 0 \leq \left(\frac{2\beta}{3} + \frac{\alpha}{3}\right)p(P_3) + \left(\frac{2\beta}{3} + \frac{\gamma}{3}\right)p(\overline{P}_3) + \gamma p(\overline{K}_3)$$

$$\begin{aligned} p(K_2) &= \frac{2}{3}p(P_3) + \frac{1}{3}p(\overline{P}_3) \\ &\leq \left(\frac{2}{3} + \frac{2\beta}{3} + \frac{\alpha}{3}\right)p(P_3) + \left(\frac{1}{3} + \frac{2\beta}{3} + \frac{\gamma}{3}\right)p(\overline{P}_3) + \gamma p(\overline{K}_3) \end{aligned}$$

$$\blacktriangleright \text{Note: } p(P_3) + p(\overline{P}_3) + p(\overline{K}_3) = 1$$

SDP Formulation

$$\blacktriangleright p(K_2) \leq \left(\frac{2}{3} + \frac{2\beta}{3} + \frac{\alpha}{3}\right)p(P_3) + \left(\frac{1}{3} + \frac{2\beta}{3} + \frac{\gamma}{3}\right)p(\overline{P}_3) + \gamma p(\overline{K}_3)$$

$\blacktriangleright$ Minimise δ :

$$\blacktriangleright \frac{2}{3} + \frac{2\beta}{3} + \frac{\alpha}{3} \leq \delta$$

$$\blacktriangleright \frac{1}{3} + \frac{2\beta}{3} + \frac{\gamma}{3} \leq \delta$$

$$\blacktriangleright \gamma \leq \delta$$

$$\blacktriangleright A = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix} \succeq 0$$

$$\blacktriangleright \text{Solution: } \delta = 1/2 \text{ and } A = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}$$

$$\blacktriangleright p(K_2) \leq \frac{1}{2} - \frac{1}{3} p(\overline{P}_3)$$

$$\blacktriangleright \text{Asymptotic result: } \text{ex}(n, K_3) \leq \left(\frac{1}{2} + o(1)\right) \binom{n}{2}$$

Interpreting the Solution

- ▶ Recall $A = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}$
- ▶ $\mathbf{v}_x \mathbf{A} \mathbf{v}_x^T = \frac{1}{2(n-1)^2} (d(x) - \bar{d}(x))^2$
- ▶ **Alternative proof:**
 - ▶ $0 \leq \sum_x (d(x) - \bar{d}(x))^2$
 - ▶ Expand and re-group terms

$$0 \leq \left\| \begin{pmatrix} \circ & \circ \\ \cdot & \cdot \end{pmatrix} \right\|^2 = 1 - 2K_2 - \frac{2}{3} \bar{P}_3$$

- ▶ Algebra structure (in limit)

Further Ways of Obtaining Inequalities

► Recall:

- $0 \leq \mathbb{E}_x(\mathbf{v}_x \mathbf{A} \mathbf{v}_x^T)$
- $\mathbf{v}_x = (\frac{d(x)}{n-1}, \frac{\bar{d}(x)}{n-1})$, $x \in V(G)$

► Extensions:

- $\mathbf{v}_x$: densities of **k-vertex** graphs rooted at x
- $\mathbf{v}_{xy}$: k -vertex graphs rooted at **edge** xy
 - Sum over $xy \in E(G)$
- $\mathbf{v}_{xy}$: k -vertex graphs rooted at **non-edge** xy
- ...

Semi-definite program (SDP)

- ▶ Data for Turán problem for K_3 :
 - ▶ Stay within 3-vertex graphs
 - ▶ Unlabelled graphs $\overline{K}_3, \overline{P}_3, P_3$
 - ▶ Objective function $0 \overline{K}_3 + \frac{1}{3} \overline{P}_3 + \frac{2}{3} P_3$
 - ▶ Use only type $\bullet$
 - ▶ Multiplication coefficients for $\bullet$ -flags
- ▶ **Primal SDP:**
 - ▶ Fix N
 - ▶ $\mathcal{F} := \{\text{unrooted graphs with } N \text{ vertices}\}$
 - ▶ d : coefficients of the objective function
 - ▶ $\mathcal{T} := \{\text{suitable types}\}$
 - ▶ $\mathcal{M}^\tau$: multiplication coefficients for τ -flags

$$\begin{aligned} \min_{X, c, \delta} \quad & \delta \\ \text{s.t.} \quad & \forall F \in \mathcal{F} : \sum_{\tau \in \mathcal{T}} M_F^\tau \cdot X^\tau + c_F = \delta - d_F \\ & \forall \tau \in \mathcal{T} : X^\tau \succeq 0 \\ & \forall F \in \mathcal{F} : c_F \geq 0 \end{aligned}$$

Dual SDP

► Dual SDP

$$\begin{array}{ll}\max_{\mathbf{y}} & \mathbf{d} \cdot \mathbf{y} \\ \text{s.t.} & \forall \mathcal{T} \in \mathcal{T} : \sum_F y_F M_F^T = Z^T \\ & \forall \mathcal{T} \in \mathcal{T} : Z^T \succeq 0 \\ & \forall F \in \mathcal{F} : y_F \geq 0 \\ & \sum_F y_F = 1\end{array}$$

- $\mathbf{y}$ describes how an optimum could look like

Structure of (Almost) Extremal Graphs

- ▶ $p(K_2) \leq \frac{1}{2} - \frac{1}{3} p(\overline{P}_3)$
- ▶ $p(K_2) \approx \frac{1}{2} \Rightarrow p(\overline{P}_3) = o(1)$
- ▶ Typical vertex x (root)
- ▶ Parts $A := \Gamma(x)$ and $B := \overline{\Gamma}(x)$
- ▶ A independent
- ▶ Few $\overline{P}_3$ via x
 - ▶ B almost independent
 - ▶ $G[A, B]$ almost complete
- ▶ G is almost $K_{A,B}$
- ▶ $p(K_2) \approx \frac{1}{2} \Rightarrow$ parts almost equal
- ▶ **Stability (Erdős'67, Simonovits'68):**
 $\forall$ almost extremal G_n is $o(n^2)$ -close to T_n^2
- ▶ **More work:** exact result for $n \geq n_0$
- ▶ **P.-Sliachan-Tyros'19:** sufficient criteria

Packages for generating flag algebra proofs

- ▶ **Rahil Baber:** `DensityBouncer`
- ▶ **Emil Vaughan:** `Flagmatic`
- ▶ **Jakub Sliačan:** `Permpack`
- ▶ **Leonardo Coregliano:** `flag-algebra-program-package`
- ▶ **Rémi de Verclos:** `rust-flag-algebras`
- ▶ **Daniel Brosch:** `FlagSOS.jl`
- ▶ **Levente Bodnár:** `FlagAlgebraToolbox`

FlagAlgebraToolbox

- ▶ ChatGPT 5.6: *“This is my default recommendation”*
- ▶ **Bodnár:** *“FlagAlgebraToolbox: Flag Algebra Computations in SageMath”*, arXiv:2601.06590
- ▶ Installs on Linux and Windows Subsystem for Linux (WSL)
 - ▶ Docker (in progress)
- ▶ Turán problem for K_3 :

```
G = GraphTheory

triangle = G(3, edges=[[0,1],[0,2],[1,2]])
edge = G(2, edges=[[0,1]])

G.exclude(triangle)
G.optimize(edge, 3, maximize=True, exact=True)
```

Inducibility

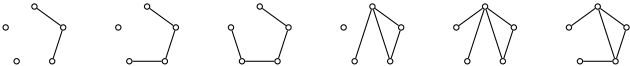
- ▶ Pippenger-Golumbic'75: $i_F := \max\{p(F, W) : \text{graphon } W\}$
- ▶ Pippenger-Golumbic'75, Bollobás-Nara-Tachibana'86, Exoo'86, Brown-Sidorenko'94, Hirst'14: i_F for $v(F) \leq 4$ except P_4
- ▶ Pippenger-Golumbic'75, Brown-Sidorenko'94, Even-Zohar & Linial'15, Balogh-Hu-Lidický-Pfender'16, P.-Sliačan-Tyros'19, Liu-Mubayi-Reiher'23, Liu-P.-Sharifzadeh-Staden'23: 9 out of 18 pairs $F, \bar{F}$ with $v(F) = 5$
 - ▶ 34 non-isomorphic F with 2 self-complimentary
- ▶ **Fractalizer**: Extremal graphon is balanced iterated blowup of F
- ▶ Balogh-Hu-Lidický-Pfender'16: C_5 is a fractalizer

New results for 5-vertex F by FlagAlgebraToolbox

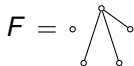
► Bodnár-P'26:

Index	F	W	i_F	part ratios
9			$\frac{5}{24}$	$\frac{3-\sqrt{3}}{12}, \frac{3-\sqrt{3}}{12}, \frac{3+\sqrt{3}}{12}, \frac{3+\sqrt{3}}{12}$
10			$\frac{5}{32}$	$\frac{3-\sqrt{3}}{12}, \frac{3-\sqrt{3}}{12}, \frac{3+\sqrt{3}}{12}, \frac{3+\sqrt{3}}{12}$
14			$\frac{15625}{62208}$	$\frac{1}{2}, \frac{1}{2}$ ($\frac{5}{6}$ -quasirandom)

► Jain-Michelen-Wei' ≥ 26 : $\forall p \in (0, 1)$ $W \equiv p$ is never extremal

► Open: 

Sample script of FlagAlgebraToolbox



```
G = GraphTheory
F = G(5,edges=[[0,1],[0,2],[0,3]])

R = QQ[sqrt(3)]; sq3 = R(sqrt(3))
r1 = 1/4+sq3/12
r2 = 1/4-sq3/12
W = G.blowup_construction(7, [r1, r1, r2, r2],
    edges=[[0, 1], [2, 3]])

lb = W.density(F)

ub = G.optimize(F, 7, exact=True, construction=W, file="cert")
```

Inducibility for 6-vertex F

- ▶ 78 complementary pairs $F, \overline{F}$
- ▶ **Brown-Sidorenko'94:** F complete partite $\Rightarrow \exists$ complete partite extremal W
- ▶ **Brown-Sidorenko'94:** F complete bipartite $\Rightarrow \exists$ complete bipartite extremal W

Previously known cases with $v(F) = 6$

	F	W	i_F	Authors
0			1	
1			$\frac{178200}{371293}$	Liu-Mubayi-Reiher'23
3			$\frac{25}{72}$	Yuster'26
8			$\frac{10}{81}$	Bollobás-Egawa-Harris-Jin'95
32			$\frac{-28+10\sqrt{10}}{9}$	Brown-Sidorenko'94

43			$\frac{24}{1555}$	Blumenthal-Phillips'25
51			$\frac{24}{1555}$	Brandt'16
53			$\frac{5}{16}$	Pippenger-Golumbic'75
56			$\frac{15}{32}$	Brown-Sidorenko'94

Bodnár-Gao-León-Liu-Sun' ≥ 26 : 'plain' optimisation of FlagAlgebraToolbox can solve 6 cases from the 9 above and the following ones:

	F	W	i_F	Remarks
2			$x^3 + \frac{7311}{10}x^2 - \frac{26568}{125}x + \frac{216}{25}$	
4			$\frac{40}{243}$	
6			$\frac{5}{16}$	union of cliques
7			$\frac{160}{729}$	

12			$\frac{40}{243}$	
13			$\frac{11360-2720\sqrt{17}}{2187}$	$\frac{7-\sqrt{17}}{6}$ -quasirandom
14			$\frac{10}{81}$	
15			$\frac{45}{256}$	
17			$\frac{40}{81}$	union of cliques
21			$\frac{128}{675}$	

24			$\frac{5}{54}$	
27			$\frac{135}{1024}$	
28			$\frac{15}{64}$	
31			$\frac{10}{81}$	
36			$\frac{-70+25\sqrt{10}}{36}$	
37			$\frac{80}{243}$	union of cliques

38			$\frac{72}{625}$	
49			$\frac{5}{54}$	
50			$\frac{5}{54}$	
55			$\frac{-672+240\sqrt{10}}{625}$	
58			$\frac{45}{512}$	
64			$\frac{5}{108}$	

65			$\frac{-70+25\sqrt{10}}{81}$	
68			$\frac{72}{625}$	
69			$\frac{72}{625}$	
70			$\frac{12353145}{67108864}$	$\frac{7}{8}$ -quasirandom
71			$\frac{45}{512}$	
72			$\frac{5}{54}$	

75			$\frac{4117715}{86093442}$	$\frac{7}{9}$ -quasirandom
77			$\frac{5242880}{43046721}$	$\frac{8}{9}$ -quasirandom

Fractalizers

- ▶ Fox-Huang-Lee'15, Yuster'19: almost every graph of order $\kappa \rightarrow \infty$ is a fractalizer
 - ▶ 5-vertex F : 1 out of 18 cases
 - ▶ 6-vertex F : 2 or 3 out of 78 cases

Exact results

- ▶ $p(F, n) := \max\{p(F, G) : v(G) = n\}$
- ▶ $\delta_{\text{edit}}(G, \mathcal{P})$: normalised edit distance
- ▶ **Perfect W -stability** (0/1-valued W): $\exists C > 0 \forall G$ on $n \geq C$ vertices

$$\delta_{\text{edit}}(G, \{\text{blowups of } W\}) \leq C(p(F, n) - p(F, G))$$

- ▶ Perfect stability **implies**
 - ▶ classical stability (**Erdős'68, Simonovits'68**)
 - ▶ exact result ($\forall$ large extremal G is a blowup of W)
- ▶ **Füredi'15, Norin-Yepremyan'17,18, P.-Sliachan-Tyros'19**
- ▶ **Bodnár-P'26, Bodnár-Gao-León-Liu-Shumin' ≥ 26 :**
 - ▶ Perfect stability for each solved F with $\{0, 1\}$ -valued W
 - ▶ Sufficient criteria of **P.-Sliachan-Tyros'19**
 - ▶ Parts of verification automated in FlagAlgebraToolbox
 - ▶ Uniqueness of W in all cases

Conjectures for $v(F) = 6$

	F	W	Density	Comments
9			0.16772113	
11			0.12658311	quasirandom edge
19			0.08738191	splitting field of a degree-221 polynomial
20			0.05432069	
22			0.06173105	

23			0.11162125	found by ChatGPT 5.5
29			0.01553174	conjectured fractalizer
34			0.06503110	
57			0.12978727	quasirandom edge
62			0.06503111	
67			0.03908943	two quasirandom edges, found by ChatGPT 5.5

76			0.14908545	
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Clique Minimisation Problem

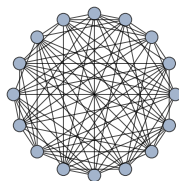
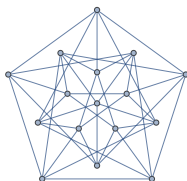
- ▶ Erdős'62:

$$f(n, k, \ell) := \min\{\#(K_k, G) : \alpha(G) < \ell, v(G) = n\}$$

- ▶ $c_{k,\ell} := \lim_{n \rightarrow \infty} f(n, k, \ell) / \binom{n}{k} = \min\{p(K_k, W) : \overline{K}_\ell\text{-free } W\}$
- ▶ **Motivation:** Ramsey numbers, Ramsey multiplicity
- ▶ Goodman'56, Lorden'62, Nikiforov'01

Results via flag algebras

- ▶ **Das-Huang-Ma-Naves-Sudakov'13, P.-Vaughan'13:**
 - ▶ $c_{3,\ell} = (\ell - 1)^{-2}$ for $4 \leq \ell \leq 7$ (union of $\ell - 1$ cliques)
 - ▶ $c_{4,3} = \frac{3}{25}$, $c_{5,3} = \frac{31}{625}$ (C_5 with loops)
- ▶ **Parczyk-Pokutta-Spiegel-Szabó'23:** $c_{4,5} = \frac{26}{2197}$ (Cayley graph on $\mathbb{Z}_{13}$ wrt cubic non-residues)
- ▶ **P.-Vaughan'13:** $c_{6,3} = \frac{19211}{1048576}$, $c_{7,3} = \frac{98491}{16777216}$
- ▶ **Clebsch graph L :**
 - ▶ $V = \{\mathbf{X} \in \{0, 1\}^5 : \text{dist}(\mathbf{X}, \mathbf{0}) \text{ is even}\}$
 - ▶ $\mathbf{X} \sim \mathbf{Y}$ iff $\text{dist}(\mathbf{X}, \mathbf{Y}) = 4$
- ▶ L and $\bar{L}$:



- ▶ **Bodnár-P' ≥ 26 :** $c_{8,3} = \frac{491411}{268435456}$

The code for minimising K_8 -density

```
G = GraphTheory
G.exclude(G(3))

clebsch_nonedges = ...

N=8; W = G.blowup_construction(N, 16, edges=clebsch_nonedges)

K8 = G(8, edges=list(itertools.combinations(range(8), 2)))

G.solve_sdp(K8, N, file="er83",
             construction=W, maximize=False,
             solver="/usr/local/bin/sdpa_qd")

G.round_solution("er83", "er83_cert",
                 denom=2**20, kernel_denom=2**15, kernel_threshold=1e-5)
```

Karush-Kuhn-Tucker conditions

- ▶ **KKT**: necessary conditions satisfied by an optimal primal-dual pair with zero duality gap

$$u = \mathbf{d}^t \cdot \mathbf{y}$$

$$Z^\tau \cdot X^\tau = 0$$

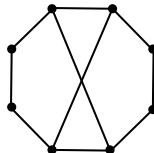
$$\mathbf{c} \cdot \mathbf{y} = 0$$

- ▶ Combinatorial interpretation: the squares must be zero on the construction

An open problem

► Conjecture (P.-Vaughan'13):

$$c_{4,4} = \frac{-11 + 14 \cdot 2^{1/3}}{192}$$



Turán problem for 3-graphs

- ▶ $ex(n, F) := \max\{|G| : G \subseteq \binom{[n]}{3}, F\text{-free}\}$
- ▶ $\pi(F) := \lim_{n \rightarrow \infty} ex(n, F) / \binom{n}{3}$
- ▶ $C_5^- := \{abc, bcd, cde, dea\}$
- ▶ Mubayi-P.-Sudakov'11: Is $\pi(C_5^-) = 1/4$?
- ▶ Bodnár-León-Liu-P.' ≥ 26 , Lidický-Mattes-Pfender' ≥ 26 : Yes!
- ▶ Final step:
 - ▶ **max-cut** vertex partition $V(G) = V_0 \cup V_1 \cup V_2$
 - ▶ $|G[V_0, V_1, V_2]| \geq (n/3)^3 - cn^3$
 - ▶ $B := \{2 + 1 \text{ edges in } G\}$
 - ▶ $M := \{1 + 1 + 1 \text{ edges not in } G\}$
 - ▶ **Aim:** $|B| \leq |M|$

```
G0 = Theory("ThreePartiteGraph", arity=2, relation_name="edges0")
G0.exclude(cocherry, K4)
CombinedTheory = combine("TwoThreeGraph", ThreeGraphTheory, G0)
```

Assumptions

► Adding assumptions:

```
G=GraphTheory
triangle=G(3,edges=[[0,1],[0,2],[1,2]])
edge=G(2,edges=[[0,1]])

W=G.blowup_construction(4,[1/sq2, 1-1/sq2],edges=[[0,0]])

G.optimize(triangle, 4, positives=[1/2-edge],
           construction=W, exact=True)
```

Thank you!

For **much more**: L.Bodnár: *"FlagAlgebraToolbox: Flag Algebra Computations in SageMath"*, arXiv:2601.06590